

Shravan Balaji

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EDUCATION

University of California, Berkeley

Berkeley, CA

Bachelor of Arts, Computer Science and Neuroscience (CS GPA: 3.93/4.00)

Expected Graduation: May 2027

Honors: Dean's List

Relevant Coursework: Discrete Mathematics, Probability Theory, Multivariable Calculus, Linear Algebra, Differential Equations, Machine Learning, Artificial Intelligence, Optimization Models, Computer Architecture

EXPERIENCE

Millennium Management

Miami, FL

Quantitative Developer Intern

June 2026 – Present

- ML Research for Systematic Data Platform (Summer 2026)

Berkeley Brain-AI Resilience Center

Berkeley, CA

Machine Learning Researcher

September 2025 – Present

- Developing supervised pose estimation models using **DeepLabCut** and **ResNet** architectures to extract high-precision **spatiotemporal** data from murine behavioral assays.
- Architecting an end-to-end ML pipeline integrating **Variational Autoencoders** and **Hidden Markov Models** to discover **latent behavioral motifs** and identify biomarkers for **stress resilience**.

Amazon

Palo Alto, CA

Software Development Engineer Intern

May 2025 – August 2025

- Optimized feature processing pipeline for Amazon's **real-time search** ranking engine, reducing latency by ~**60%** and improving data freshness, enabling **faster feature computation** across millions of daily queries
- Developed a data ingestion and attribution pipeline leveraging **Scala** and **Apache Spark**, deployed via **AWS Lambda** to **DynamoDB**, accomplishing **sub-second** delivered latency on average
- Implemented and maintained cloud infrastructure on **AWS** using **CDK** for automated and scalable deployment; **automated CI/CD pipeline** with JUnit test suites, ensuring at least **>90% code coverage**

Berkeley Seismological Laboratory

Berkeley, CA

Machine Learning Engineer

January 2025 – Present

- Developed and deployed **CNN classifier models** in **PyTorch** to automate real-time anomaly detection and filter instrument noise, improving core seismic signal quality by **11%** with **>99% classification accuracy**
- Implemented a high-dimensional **DBSCAN** clustering algorithm to segment over **104,000 power spectral density (PSD)** frequency distribution plots, identifying **latent structural patterns** and data anomalies
- Engineered an automated system for seismic data visualization using **Matplotlib** and **Seaborn**, boosting analysis **efficiency by 30%** and enabling real-time monitoring

UCSF Health

San Francisco, CA

Machine Learning Researcher

November 2024 – August 2025

- Trained and evaluated **Random Forest classification models** on multi-channel sleep **EEG/EMG data** from murine behavioral experiments using **Scikit-Learn** and **DeepLabCut**

PROJECTS

Voice Trainer | [Github](#) | *Python, Vosk, PyTorch, Numpy* | 1st place in Hiya VoiceAI Hackathon

Berkeley, CA

- Built a real-time audio processing engine using **NumPy** to extract and analyze speech features: pacing (WPM), volume (RMS), and pitch variation (ZCR), delivering sub-100ms feedback latency
- Designed a modular, high-throughput analysis pipeline that seamlessly integrated the Vosk API for **offline speech recognition** alongside OpenAI Whisper for deep post-session structural transcriptions

WordNet: Semantic Lexicon and Graph Analysis Tool | *Java, Git*

Berkeley, CA

- Designed a Java-based semantic lexicon using graph data structures to model word relationships/meanings
- Executed comprehensive JUnit testing procedures to ensure at least 95% code reliability during assessments

SKILLS

Programming/Scripting Languages: Java, Python, Scala, SQL, C, C++, MATLAB, Bash, RISC-V, TypeScript

Cloud/Data Frameworks: AWS (S3, Lambda, SQS, SNS, DDB), Apache Spark, PostgreSQL, MongoDB

Data Science and Analytics: Pandas, Scikit-Learn, PyTorch, TensorFlow, HuggingFace, Numpy

ACHIEVEMENTS

Citadel Collegiate Quant League (**Top 15 National**); **AIME Qualifier** (x2); USNCO National Qualifier (**Top 200**); USABO **Top 225 National**; NASA TechRise Winner (**Team Leader**); Delaware Math League **1st Place** (x2), DelawareLaunch Biotech **Pitch Competition 1st Place**, **1590 SAT**, **36 ACT**